

Zhengqin Shu

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EDUCATION

Tsinghua University, Xingjian College, Tsien Excellence in Engineering Program Sept. 2022-Present
Major in Theoretical and Applied Mechanics, Minor in Software Engineering.

- **Overall GPA: 3.98/4.0; Rank: 1/28.**

- **Relevant Coursework:**

Basics of Computational Fluid Dynamics	A+	Foundation of Solid Mechanics	A+
Viscous Fluid Mechanics	A+	Lab. of Fundamental Mechanics	A+
Thermodynamics and Statistical Physics	A+	Introduction to Artificial Intelligence	A+

PUBLICATIONS

- **Z. Shu** and C. Xu, “Mean velocity profiles and total shear stress profiles in adverse-pressure-gradient turbulent boundary layers considering history effect”, Submitted to *J. Fluid Mech.*, 2025, arXiv: arxiv.org/abs/2508.07663.
- Y. Li, **Z. Shu**, W. Huang, and C. Xu, “Inflow turbulence generation for turbulent boundary layer with adverse pressure gradient using an equivalent boundary layer model”, *Phys. Fluids*, 2025, doi.org/10.1063/5.0296278.
- **Z. Shu**, D. Zhou, and H. J. Bae, “Analogous dynamics of Couette-Poiseuille flows and adverse-pressure-gradient turbulent boundary layers”, manuscript in preparation.
- Y. Liu, **Z. Shu**, and C. Xu, “Structures and transport process in turbulent channel flows based on triple decomposition”, manuscript in preparation.

CONFERENCE

- **Z. Shu**, D. Zhou, and H. J. Bae, “Analogous dynamics of Couette-Poiseuille flows and adverse-pressure-gradient turbulent boundary layers”, American Physical Society 78th Annual Meeting of the Division of Fluid Dynamics (**APS DFD 2025 Meeting**), Houston, U.S., Nov. 2025.
- Y. Li, **Z. Shu**, W. Huang, and C. Xu, “Inflow turbulence generation for adverse-pressure-gradient turbulent-boundary-layers based on equivalent boundary layer model”, the 31st International Conference on Computational & Experimental Engineering and Sciences (ICCES 2025), Changsha, China, May 2025.

HONORS AND AWARDS

- APS Annual Meeting of the Division of Fluid Dynamics (APS-DFD) Travel Award Nov. 2025
- **Gold Medal** in the 38th Chinese Physics Olympiad. (CPhO) Dec. 2021
- **National Scholarship** (Ministry of Education of China, 5 out of 147) Oct. 2024
- Top 6/620 in the 4th International Engineering Mechanics Contest, Asian Region. (**Special Prize**) Dec. 2023
- First Prize in the 14th Chinese College Mathematics Competition (Non-mathematics Category) Jan. 2023
- Tsinghua University Scholarship for Excellence in All Aspects 2023, 2024, 2025

- ShuPing Scholarship (including 3 special prizes)

2020-2024

RESEARCH EXPERIENCE

California Institute of Technology, Graduate Aerospace Laboratories

Pasadena, U.S.

Advisor: Prof. H. Jane Bae

Jul. 2025-Present

Project: Analogous dynamics of CP flows and APG TBLs

- Observed the inner-region similarity of Couette-Poiseuille (CP) flows and adverse-pressure-gradient (APG) turbulent boundary layers (TBLs) at an approximately ten-time larger friction Reynolds number, laying the foundation for CP-flow-based APG TBL modelling.
- Modified the mixing length model with pressure gradient, and proposed a novel wall model for wall-bounded turbulence with APGs.

Tsinghua University, School of Aerospace Engineering

Beijing, China

Advisor: Prof. Chunxiao Xu

Mar. 2024-Present

Project1: Analysis of APG TBLs considering history effect

- Introduced a new scaling for turbulent boundary layers with adverse pressure gradients, and by employing an estimation-correction model, obtained accurate velocity and stress profiles. For the first time, history effect was explicitly characterized.
- Applied the shear stress model of APG TBLs to the equivalent boundary layer (EBL) model for inflow turbulence generation, accurately reproducing turbulence fluctuations and enabling realistic downstream development in simulations.
- Currently trying to extend the original estimation-correction model into a numerically reproducible iterative method by introducing a virtual profile.

Project2: Analysis of structures and transport process based on triple decomposition

- Studied the profile of mean velocity, Reynolds stress, and the transport process based on the triple decomposition, laying the foundation for developing a numerical model.
- Elucidated the transport process of turbulent kinetic energy among the mean flow, coherent fluctuations, and background fluctuations, and emphasized the influence of vortex inclination on the redistribution term.

EXTRA-CURRICULAR ACTIVITIES

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| • President of Student Magic Club of Tsinghua University (of over 400 members) | 2023-2024 |
| • Executive director of the 5 th Tsinghua University Magic Gala | Dec. 2023 |
| • Chief stage manager of the 6 th Tsinghua University Magic Gala | Nov. 2024 |
| • 10 individual magic shows performed with audience over 200 | 2022-2025 |
| • 2 individual original magic shows performed with audience over 2000 | 2023 |
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TECHNIQUES AND SKILLS

Programming: C++, Python, Matlab, Fortran.

Software: Solidworks, AutoCAD, Labview, LaTeX.

Operating Systems: familiar with Linux/Unix, High-Performance Computing (HPC) Environment.

Skills: Particle Imaging Velocimetry (PIV), Hot Wire measurement, Mesh generation.